

A CUBE-NULL CONTRAPOSITIVE FOR THE ARONSZAJN-NULL GAME

CANDIDATE SOLUTION PACKET FOR ARXIV:MATH/0601556

STATUS

Candidate full solution, likely valid, for human review. The packet proves the contrapositive

Player I wins the Aronszajn-null game on $A \implies A$ is not cube-null.

For Borel subsets of separable real Banach spaces, Csornyei’s theorem identifies cube-null and Aronszajn-null sets. Thus this gives an affirmative answer to Duda–Tsaban’s Conjecture 2.4.

This is the single canonical packet for arXiv:math/0601556 in the run. The earlier finite-dimensional and cylindrical-pullback partial packets are retained inside this packet as superseded provenance only; they are no longer separate current solution packets for this target.

SOURCE QUESTION

The source paper is Jakub Duda and Boaz Tsaban, *Null sets and games in Banach spaces*, arXiv:math/0601556, later published in *Topology and its Applications* 156 (2008), 56–60.

Duda and Tsaban define, for a nonzero vector $v \in X$, $\mathcal{A}(v)$ to be the family of Borel sets whose intersection with every affine line parallel to v has one-dimensional Lebesgue measure zero. A Borel set $A \subset X$ is Aronszajn-null if for every dense sequence (v_n) in X there are $A_n \in \mathcal{A}(v_n)$ with $A \subset \bigcup_n A_n$.

In the game $\mathsf{G}_{\mathcal{A}}(A)$, Player I plays directions v_n so that the resulting sequence is dense in X , Player II answers with $B_n \in \mathcal{A}(v_n)$, and Player II wins if $A \subset \bigcup_n B_n$. The authors prove that if Player I has no winning strategy, then A is Aronszajn-null, and ask for the converse.

For a game G , the notation $\mathbf{I} \uparrow \mathsf{G}$ is a shorthand for “ $\mathbf{I}$ has a winning strategy in the game G ”, and $\mathbf{I} \nmid \mathsf{G}$ stands for “ $\mathbf{I}$ does not have a winning strategy in the game G ”. Define $\mathbf{II} \uparrow \mathsf{G}$ and $\mathbf{II} \nmid \mathsf{G}$ similarly. The following is easy to see.

Lemma 2.3. *If $\mathbf{I} \nmid \mathcal{A}_{\mathsf{G}}$ for A , then A is Aronszajn-null.* □

The converse is open.

Conjecture 2.4. *If A is Aronszajn-null, then $\mathbf{I} \nmid \mathcal{A}_{\mathsf{G}}$ for A .*

Lemma 2.5. *The property $\mathbf{II} \uparrow \mathcal{A}_{\mathsf{G}}$ is preserved under taking Borel subsets and countable unions, i.e., it defines a Borel σ -ideal.*

Proof. It is obvious that $\mathbf{II} \uparrow \mathcal{A}_{\mathsf{G}}$ is preserved under taking Borel sub-

PROOF INTUITION

The missing point in the source conjecture is adaptivity: Aronszajn-nullness defeats every fixed dense direction sequence, while a Player I strategy chooses directions after seeing Player II’s previous null sets. The proof below turns this adaptivity against Player I.

Against a supposed winning strategy for Player I, Player II makes one special legal play. At inning n , Player II does not choose one prospective Csornyei cube-construction obstruction;

instead, Player II plays the countable union of all bad sets associated with all finite rational cube prefixes already available. Each bad set is null on the current family of parallel lines, so the countable union is a legal move. Since Player I's strategy is assumed to win, there is a point $a \in A$ outside all these bad sets. Being outside all bad sets lets a thread an infinite sequence of rational cube widths and Borel approximation indices. The resulting translated cube measure gives positive mass to A , so A is not cube-null.

PRELIMINARIES

We use the following standard descriptive-set-theoretic decomposition of a Borel set. It is the same device used in Csornyei's cube-measure construction and in later streamlined presentations of the Aronszajn-null/cube-null equivalence.

Lemma 1 (Borel approximation tree). *Let E be a Borel subset of a Polish space. There are Borel sets*

$$E_s \quad (s \in \mathbb{N}^{<\mathbb{N}})$$

such that:

- (1) $E_\emptyset = E$;
- (2) $E_{s0} \subset E_{s1} \subset \cdots \subset E_s$ and $E_s = \bigcup_{k=0}^{\infty} E_{s \smallfrown k}$ for every finite sequence s ;
- (3) for every $\alpha \in \mathbb{N}^{\mathbb{N}}$,

$$\bigcap_{n=0}^{\infty} \overline{E_{\alpha \upharpoonright n}} \subset E.$$

Proof. Realize E as the injective continuous image $\phi(C)$ of a closed subspace C of the Baire space $\mathbb{N}^{\mathbb{N}}$. For $s = (s_1, \dots, s_m)$, let

$$Z_s = \{\alpha \in \mathbb{N}^{\mathbb{N}} : \alpha_i \leq s_i \text{ for } 1 \leq i \leq m\}, \quad E_s = \phi(C \cap Z_s).$$

The first two assertions are immediate. For the last, if $x \in \bigcap_m \overline{E_{\alpha \upharpoonright m}}$, choose $x_m = \phi(\beta_m) \in E_{\alpha \upharpoonright m}$ with $x_m \rightarrow x$. Replacing the tail of β_m by zero puts the sequence in the compact product $\prod_i \{0, \dots, \alpha_i\}$, so a subsequence converges pointwise to some $\beta \in C$. Continuity gives $x = \phi(\beta) \in E$. $\square$

For a nonzero $u \in X$ and $q > 0$, write $\lambda_{u,q}$ for the uniform probability measure on the segment $\{tu : -q \leq t \leq q\}$.

Lemma 2 (One-step bad sets are legal). *Let E_s be as in Lemma 1. Fix a finite sequence s , a Borel probability measure μ on X , a nonzero vector u , a constant $0 < c < 1$, and a number $\rho > 0$. Define*

$$G = \{y \in X : (\delta_y * \mu)(E_s) > c\}$$

and

$$B = G \setminus \bigcup_{\substack{k \in \mathbb{N} \\ q \in \mathbb{Q}, 0 < q < \rho}} \{y \in X : (\delta_y * \lambda_{u,q} * \mu)(E_{s \smallfrown k}) > c\}.$$

Then $B \in \mathcal{A}(u)$.

Proof. The set B is Borel because $y \mapsto (\delta_y * \nu)(D)$ is Borel for each Borel D and Borel probability ν .

Assume that an affine line $L = a + \mathbb{R}u$ has $B \cap L$ of positive one-dimensional measure. Let

$$f(a + tu) = (\delta_{a+tu} * \mu)(E_s).$$

Since $f > c$ on B , there is $\eta > 0$ such that

$$D = \{t : a + tu \in B \text{ and } f(a + tu) > c + \eta\}$$

has positive measure. Pick a Lebesgue density point t_0 of D . For a sufficiently small rational $q \in (0, \rho)$,

$$\lambda\{r \in [-q, q] : t_0 + r \in D\} / (2q) > \frac{c}{c + \eta}.$$

Hence

$$\begin{aligned} (\delta_{a+t_0u} * \lambda_{u,q} * \mu)(E_s) &= \int_{-q}^q f(a + (t_0 + r)u) \frac{dr}{2q} \\ &> (c + \eta) \frac{c}{c + \eta} = c. \end{aligned}$$

Because $E_s = \bigcup_k E_{s \cap k}$ increasingly, some k satisfies

$$(\delta_{a+t_0u} * \lambda_{u,q} * \mu)(E_{s \cap k}) > c.$$

This contradicts $a + t_0u \in B$. Therefore every line parallel to u meets B in a null set. $\square$

MAIN THEOREM

Theorem 3. *Let X be a separable real Banach space and let $A \subset X$ be Borel. If Player I has a winning strategy in $\mathsf{G}_A(A)$, then A is not cube-null. Consequently, by Csornyei's theorem, if A is Aronszajn-null then Player I has no winning strategy in $\mathsf{G}_A(A)$.*

Proof. Assume that σ is a winning strategy for Player I on A . Fix $0 < c < 1$, for instance $c = 1/2$. Let A_s , $s \in \mathbb{N}^{<\mathbb{N}}$, be a Borel approximation tree for A as in Lemma 1.

We define one legal counterplay by Player II. Suppose that the first m responses $C_0, \dots, C_{m-1}$ have already been defined, and let

$$x_{m+1} = \sigma(C_0, \dots, C_{m-1})$$

be Player I's next direction. Put $u_{m+1} = x_{m+1} / \|x_{m+1}\|$. Player I's legal moves are nonzero; scaling does not change the family $\mathcal{A}(x_{m+1})$.

Let $\mathcal{P}_m$ be the countable set of all prefixes

$$p = ((q_1, k_1), \dots, (q_m, k_m))$$

with $k_i \in \mathbb{N}$ and rational $0 < q_i < 2^{-i}$. For such p , write

$$s(p) = (k_1, \dots, k_m), \quad \mu_p = \lambda_{u_m, q_m} * \dots * \lambda_{u_1, q_1},$$

with the convention that for the empty prefix $p = \emptyset$, $s(p) = \emptyset$ and $\mu_p = \delta_0$.

For each $p \in \mathcal{P}_m$, define

$$G_p = \{y : (\delta_y * \mu_p)(A_{s(p)}) > c\},$$

and

$$B_p = G_p \setminus \bigcup_{\substack{k \in \mathbb{N} \\ q \in \mathbb{Q}, 0 < q < 2^{-(m+1)}}} \{y : (\delta_y * \lambda_{u_{m+1}, q} * \mu_p)(A_{s(p) \cap k}) > c\}.$$

By Lemma 2, $B_p \in \mathcal{A}(u_{m+1}) = \mathcal{A}(x_{m+1})$. Set

$$C_m = \bigcup_{p \in \mathcal{P}_m} B_p.$$

Since $\mathcal{A}(x_{m+1})$ is closed under countable unions, C_m is a legal response for Player II.

This recursively defines a legal play against σ . Since σ is winning for Player I, the resulting direction sequence is dense and

$$A \not\subset \bigcup_{m=0}^{\infty} C_m.$$

Choose

$$a \in A \setminus \bigcup_{m=0}^{\infty} C_m.$$

We now thread a cube through a . Start with the empty prefix p_0 . Since $(\delta_a * \delta_0)(A) = 1 > c$, we have $a \in G_{p_0}$. Suppose inductively that $p_m \in \mathcal{P}_m$ and $a \in G_{p_m}$. Because $a \notin C_m$, and $B_{p_m} \subset C_m$, we have $a \notin B_{p_m}$. By the definition of B_{p_m} , there are $k_{m+1} \in \mathbb{N}$ and rational $0 < q_{m+1} < 2^{-(m+1)}$ such that, for

$$p_{m+1} = p_m \widehat{(q_{m+1}, k_{m+1})},$$

we have $a \in G_{p_{m+1}}$. Thus for every m ,

$$(1) \quad (\delta_a * \mu_{p_m})(A_{s(p_m)}) > c.$$

Let $v_i = q_i u_i$. Then $\sum_i \|v_i\| < \infty$. If T_i are independent uniform random variables on $[-1, 1]$, the series $\sum_i T_i v_i$ converges almost surely in X . Let μ be its law. Equivalently, μ_{p_m} converges weakly to μ . The span of $\{v_i : i \in \mathbb{N}\}$ is dense because it is the same linear span as $\{x_i : i \in \mathbb{N}\}$, and the actual play generated by σ is legal for Player I, so the sequence (x_i) is dense in X . Therefore $\delta_a * \mu$ is a nondegenerate cube measure.

Fix r . For all $m \geq r$, the approximation-tree nesting gives

$$A_{s(p_m)} \subset A_{s(p_r)}.$$

By (1),

$$(\delta_a * \mu_{p_m})(A_{s(p_r)}) > c \quad (m \geq r).$$

Passing to the weak limit and using the portmanteau theorem,

$$(\delta_a * \mu)(\overline{A_{s(p_r)}}) \geq c.$$

The closed sets $\overline{A_{s(p_r)}}$ are decreasing in r , and Lemma 1 gives

$$\bigcap_{r=0}^{\infty} \overline{A_{s(p_r)}} \subset A.$$

By continuity from above for probability measures,

$$(\delta_a * \mu)(A) \geq (\delta_a * \mu)\left(\bigcap_{r=0}^{\infty} \overline{A_{s(p_r)}}\right) = \lim_{r \rightarrow \infty} (\delta_a * \mu)(\overline{A_{s(p_r)}}) \geq c > 0.$$

Thus a nondegenerate cube measure gives positive mass to A . Hence A is not cube-null.

Csornyei proved that, for Borel subsets of separable real Banach spaces, Aronszajn-null and cube-null are the same ideal. Therefore an Aronszajn-null Borel set cannot admit a winning strategy for Player I. $\square$

Corollary 4. *For every Borel subset A of a separable real Banach space, Duda–Tsaban’s Conjecture 2.4 holds:*

$$A \text{ is Aronszajn-null} \implies \text{Player I has no winning strategy in } \mathsf{G}_A(A).$$

Together with Duda–Tsaban’s easy implication, this characterizes Aronszajn-null Borel sets as exactly those for which Player I has no winning strategy in the Aronszajn-null game.

VERIFICATION NOTES

The proof is non-computational. The main points for human review are:

- Lemma 2: the density-point argument must be checked carefully. The strict gap $f > c + \eta$ on a positive-measure subset is what turns almost-invariance under a short segment convolution into a contradiction.
- In the main proof, Player II's inning m move is the countable union of bad sets over every rational finite cube prefix. This is why the later point a can choose its prefix after the play has been generated.
- The actual direction sequence used in the cube measure is the one produced by the supposed winning strategy against this single legal play, so it is dense by the legality requirement on Player I.
- The Borel approximation tree is used only at the final portmanteau step, to pass from positive mass on nested approximants to positive mass on the original Borel set.

NOVELTY CHECK

Before promotion, the run indexes were searched for arXiv id '0601556', 'Aronszajn-null game', 'Duda Tsaban', 'Conjecture 2.4', 'cube null', 'Pawlikowski', and 'Rothberger'. The only local hits for this target were the two earlier lane-0 partial packets, covering the finite-dimensional case and finite-dimensional cylindrical pullbacks, plus the prior unfinished attempt.

External searches on 2026-06-15 for 'Aronszajn-null game Duda Tsaban conjecture', 'Null sets and games in Banach spaces Conjecture 2.4', 'Aronszajn-null Player I winning strategy', and related selection-principle phrases found the source paper but no later paper claiming to solve the game conjecture. The proof uses Csornyei's known equivalence of Aronszajn-null and cube-null sets, and the Borel/cube-measure construction is in the same family as Csornyei's argument and the later exposition by Eldredge–Gordina–Le Donne–Li, but the online countable-prefix fusion against a Player I strategy appears not to be recorded in the searched sources.

HUMAN REVIEW RECOMMENDATION

Review as a candidate full solution of Duda–Tsaban Conjecture 2.4. The most important line to audit is the transition from the supposed winning strategy to the single legal Player II play $C_m = \bigcup_{p \in \mathcal{P}_m} B_p$. If that fusion is correct, the rest is a standard cube-measure compactness argument.

REFERENCES

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